

The Future of This World Must Be Derived by the World

The World Language Model Framework: Heterogeneous Judges, Complementary Blind Spots, and Reality Feedback

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This version includes the first minimal experiment. Real LLM judges were not available; language-judge results are simulated.

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Abstract

The bottleneck of autonomous scientific discovery is not only generating hypotheses but verifying large numbers of them. Version 0.1 proposed a surrogate judge between generation and real experiments and treated generator–judge error correlation as a central variable. Our first minimal experiment shows that this direction is **partly supported but needs revision**: a judge’s intrinsic discriminative ability remains the main determinant of performance, while error correlation is better treated as a diagnostic variable. More importantly, a data-driven surrogate and language-like judges do not differ mainly in overall accuracy; they exhibit **complementary blind spots**. The surrogate also collapses under distribution shift, making an explicit “I don’t know” route a prerequisite rather than an optional feature.

We therefore upgrade the architecture from a single independent judge to **heterogeneous judges + uncertainty routing + reality feedback**. We retain false-acceptance/false-rejection metrics and the verification-cost model, but replace the degenerate resonance definition from v0.1 with direct error-correlation and error-type-specific blind-spot measures. The revised claim is: **a strong verification system requires discriminative ability, low failure-mode resonance, complementary blind spots, and reliable escalation under uncertainty or OOD inputs**.

1 The Problem: Generation Is Cheap, Verification Is Expensive

Large models can generate many candidate hypotheses quickly, but proposing a plausible statement is not the same as knowing whether it is true. Real experiments provide the most reliable feedback, yet they are slow, expensive, and sometimes risky. Autonomous scientific systems therefore need an intermediate verification layer that reduces a large candidate set to a smaller set worth testing in reality.

Version 0.1 argued that if a generator and its judge share the same blind spots, repeated self-critique may simply reproduce the same mistake. The first experiment does not remove this concern, but it weakens the original claim: **error independence**

matters, but it is not the only or dominant determinant of judge performance.

2 From One Good Judge to Multiple Complementary Judges

In the experiment, language-like judges and a data-driven surrogate exhibited sharply different error structures. Language-like judges were better at dimensional mistakes and obvious linearization errors; the surrogate was better at dissipation, forcing, and cubic dynamics that looked superficially plausible in symbolic form. Their overall balanced accuracies were similar, but their blind spots differed.

This suggests that the right architecture is not a search for one universal judge, but a **heterogeneous judge ensemble**. We call the complete closed loop the **World Language Model Framework (WLMF)**: the language model proposes, explains, and organizes candidates; heterogeneous judges test them through different failure modes; a router escalates uncertainty, disagreement, and OOD cases to the real world; and real outcomes feed back into both judging and generation. Here, “world” is not merely rhetorical: **the final authority must return to an observable, testable, feedback-producing external world.**

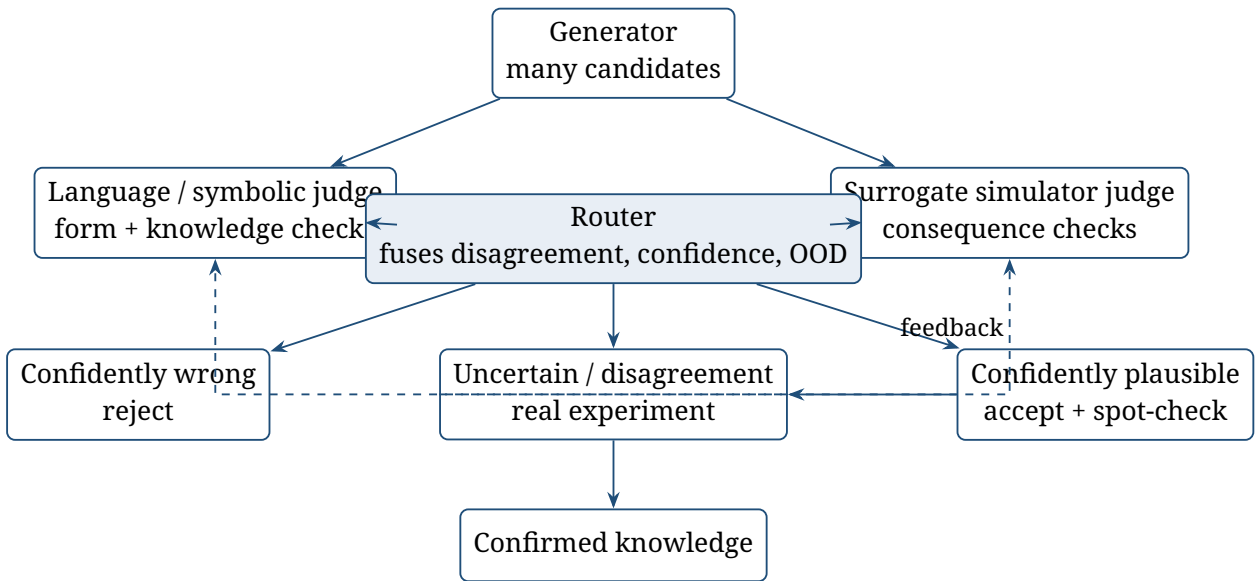


Figure 1. The World Language Model Framework (WLMF) verification loop. Different judges cover different failure modes; disagreement, low confidence, and OOD inputs are escalated to reality.

Additional judges could include theorem provers, retrieval systems, domain simulators, or physical instruments. The point is not judge count, but whether they contribute **different failure modes and partially independent evidence**.

3 How to Evaluate a Judge

3.1 Operational metrics

On the generator’s actual outputs, we measure:

- $f_p = P(\text{accept} \mid \text{wrong})$, the false-acceptance rate;
- $f_n = P(\text{reject} \mid \text{correct})$, the false-rejection rate;
- balanced accuracy $1 - (f_p + f_n)/2$.

These metrics directly determine whether a judge is operationally useful.

3.2 Diagnostic metrics: error coupling and blind spots

Version 0.1 defined

$$R_{\text{res}} = P(E_J \mid E_G) - P(E_J)$$

as an independent resonance measure. Literal execution showed that, on a single candidate stream, this quantity degenerates into a recombination of f_p and f_n and does not add independent correlation information. Version 0.2 therefore drops it as a primary metric.

Instead we use:

- $\phi = \text{corr}(\text{judge error}, \text{candidate is wrong})$ for overall error coupling;
- error-type-specific blind-spot measures that compare the excess false-acceptance rate on error class k against an overall or reference baseline.

The goal is not to prove that correlation dominates ability, but to identify **which generator failures a judge systematically misses**, so complementary judges can be selected.

4 First Minimal Experiment

4.1 Setup

We implemented the pendulum-and-spring experiment proposed in v0.1. True dynamics T^* were used only for final scoring; a surrogate $\hat{T}$ learned next-state prediction from noisy trajectories. Candidate equations included correct dynamics and several injected defects: dissipation, energy injection, periodic forcing, dimensional scaling errors, offsets, linearization, and cubic nonlinearities.

The run produced roughly 7.6×10^4 candidates and trained multiple surrogates across data sizes, seeds, and training-data contamination conditions. Because no LLM API was available in the execution environment, self/same-family/different-family language judges were **simulated from fixed symbolic feature priors**. Claims involving real LLM behavior should therefore be treated as weak evidence.

4.2 Headline results

Prediction	Outcome	Interpretation
Lower error correlation means better judges	Partly supported	Strong univariate relation, but after controlling for judge ability, correlation adds only about three percentage points of explained variance.
Surrogate clearly beats language scoring	Refuted	Overall balanced accuracy is similar; gains come from different error classes rather than higher aggregate performance.
Surrogate fails OOD	Strongly supported	Under OOD conditions the surrogate can collapse to rejecting every candidate, including all correct ones ($f_n = 1$).

Table 1. How the first experiment affected the main v0.1 predictions.

Across 54 judge points, Spearman correlation between error coupling ϕ and balanced accuracy was about -0.48 (95% CI $[-0.66, -0.26]$). But after controlling for each judge’s best attainable ability, ability alone explained about 91% of the variance, rising to about 94% when correlation was added. Thus **correlation is a useful diagnostic signal, but not the main cause of performance**.

The richer result is error-type complementarity: the surrogate is stronger on dissipation, forcing, dimensional errors, and cubic terms, while language-like judges are stronger on explicit symbolic mistakes such as linearization. This supports complementary verification rather than replacing one class of judge with another.

4.3 Causal ablation

We deliberately contaminated surrogate training trajectories to manipulate failure modes. Strong contamination moved error correlation from roughly -0.3 to $+0.4$ – $+0.8$ while balanced accuracy fell from about 0.58 to about 0.52. This provides direct evidence that **teaching a judge the same failure pattern as the generator can make the judge blind to that pattern**. Fine-grained targeted contamination was noisier across seeds and needs more repetitions.

4.4 OOD: why “I don’t know” is necessary

The surrogate produced usable long-horizon rollouts in-distribution, but its error grew by roughly 50x under larger pendulum angles and stiffer springs. In the extreme case it rejected every candidate, including all correct candidates, yielding $f_n = 1$.

A surrogate judge without reliable uncertainty or OOD detection can therefore eliminate genuine discoveries together with ordinary mistakes. The system must support at least three actions:

Accept, Reject, Escalate/Unknown.

5 Cost and Routing

Let N be the number of candidates, p the fraction that are correct, C_r the cost of a real experiment, and C_s the cost of surrogate verification. For binary filtering, the pass rate is

$$q = p(1 - f_n) + (1 - p)f_p,$$

and

$$\text{Cost}_1 = NC_s + NqC_r.$$

If missing a true discovery costs C_{miss} , the total objective becomes

$$J = NC_s + NqC_r + Npf_nC_{\text{miss}}.$$

The experiment shows that three-way routing only helps strongly when a judge's errors are concentrated in its own low-confidence region. Simulated language judges benefited more from low-confidence escalation than the surrogate, whose mistakes were not sufficiently concentrated there. Version 0.2 therefore elevates **calibrated uncertainty and an independent OOD detector** to first-class research problems.

6 Limitations and Next Steps

The current evidence remains limited:

- language judges were simulated, not real LLMs;
- the generator was a random program, not a scientific agent;
- pendulums and springs are low-dimensional ODEs and do not establish open-world scientific discovery;
- training-data contamination only approximates shared training/failure modes;
- the original resonance metric was found to be flawed under the proposed evaluation setup.

The next useful experiments are therefore modest rather than gigantic: use real LLMs from different families as judges, add an independent OOD detector, and compare a single strongest judge against a complementary multi-judge ensemble. Only after these effects replicate across tasks does scaling to larger autonomous-discovery systems become meaningful.

7 Conclusion

The first experiment does not establish that the most independent judge is the best judge. It supports a narrower and more useful conclusion: **a judge first needs sufficient discriminative ability; among capable judges, differences in failure modes determine which blind spots they can cover for one another**. A surrogate's value is not that it must be more accurate than a language judge, but that it can see errors the language judge misses, while the language judge can catch structural errors the surrogate overlooks.

The center of the proposal therefore shifts from “find an independent judge” to

heterogeneous judges + complementary blind spots + uncertainty routing + reality feedback.

A reliable autonomous discovery system may not need one judge that is always right. It needs a set of judges that **fail differently, and know when to hand the question back to reality.**

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